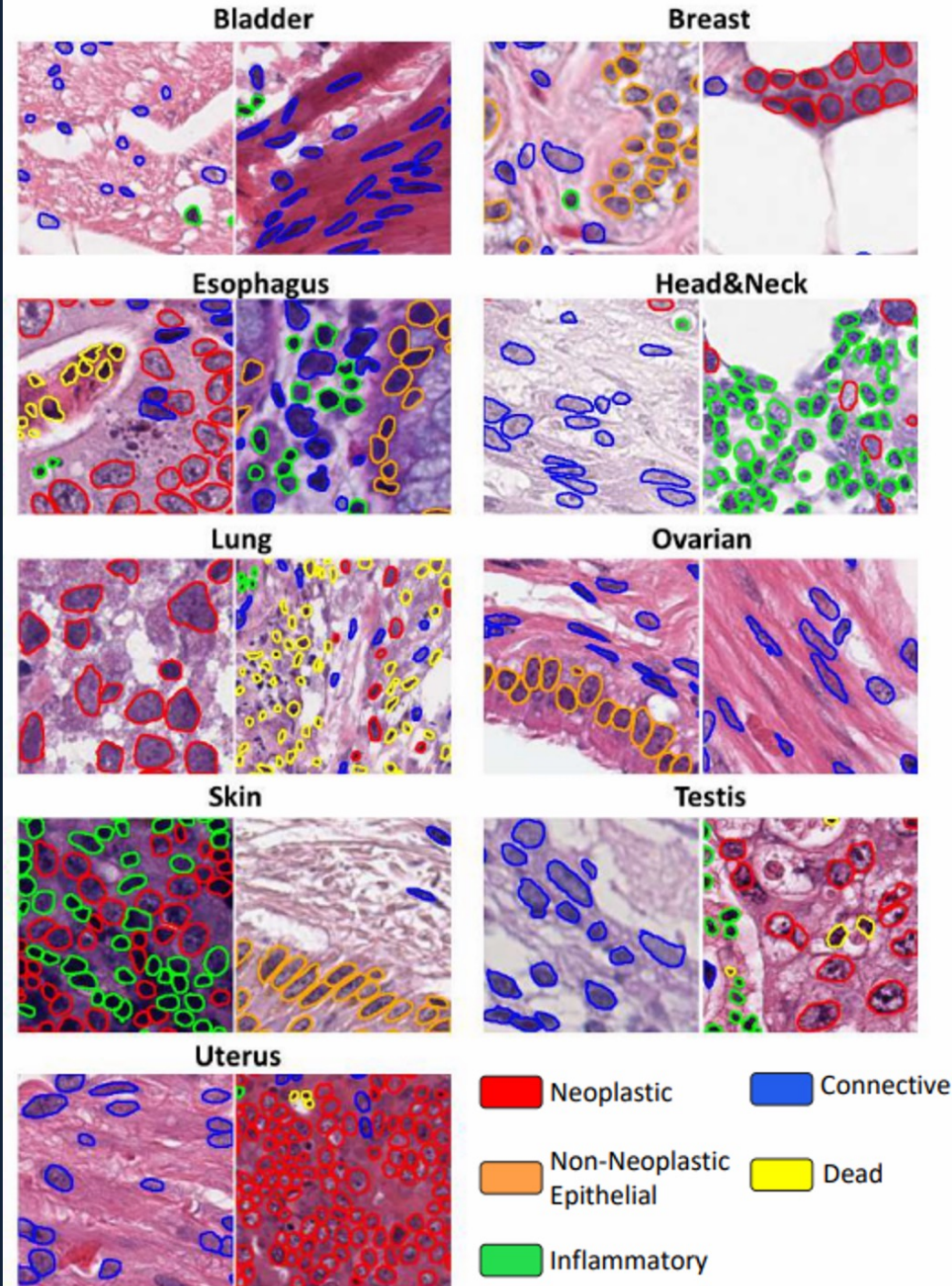


Nuclei instance segmentation across diverse tissues

HoVer-Net and CellViT-UNI

Stage I progress report
Results through September 28, 2026

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Baseline reproduction led to a detection-focused question

What limits rare-nucleus performance, and how should we measure transfer?

1

A Language priors

CONCH re-typing and output recovery did not improve the baseline.

2

B Targeted data

Copy-paste harmed Dead typing. Controlled synthesis gave no reliable detection gain.

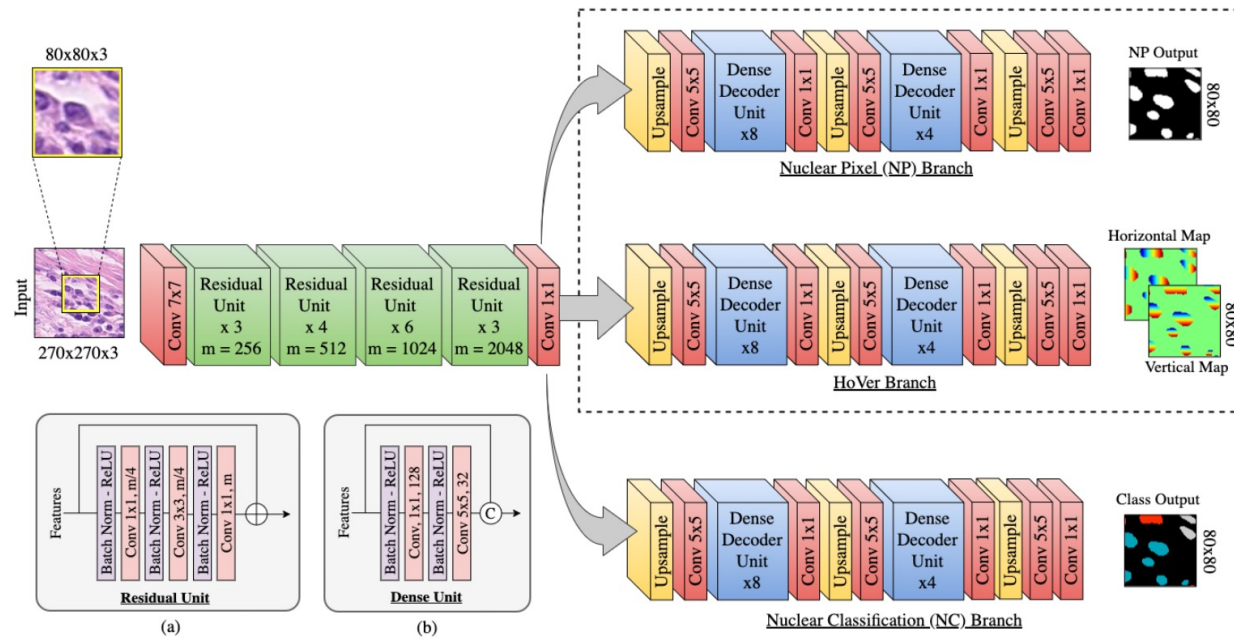
3

C Robustness evaluation

Real cross-domain transfer and appearance counterfactuals rank models differently.

Current finding: Dead nuclei are primarily lost before the type decision

A shared CNN encoder supports three prediction branches



Encoder

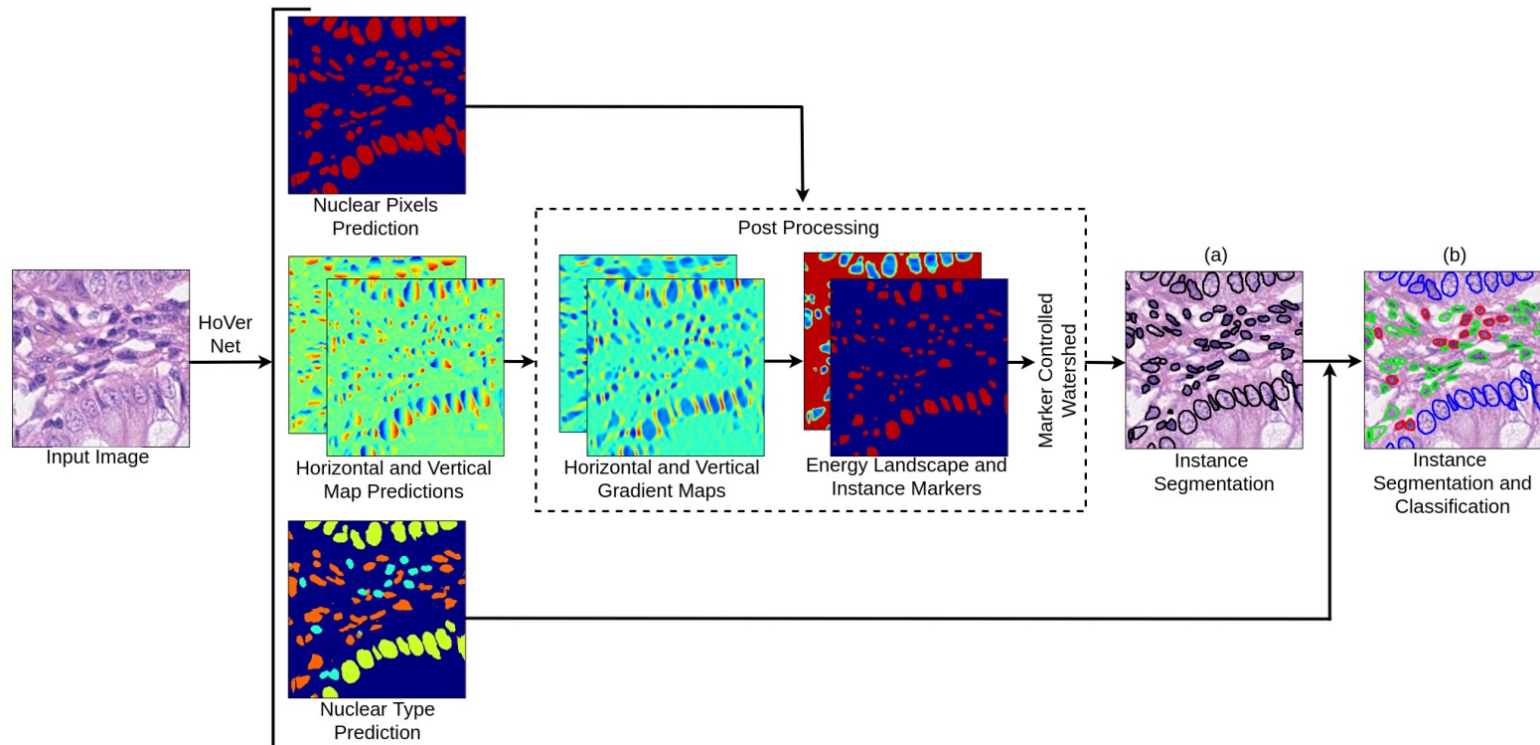
Modified Preact-ResNet50 retains a denser spatial grid for fine nuclear structure.

Decoders

Separate NP, HV and NC upsampling paths reuse encoder features through skip connections.

Original architecture figure. Paper input/output sizes describe that version, not our patch protocol.

HV geometry supplies boundaries for watershed decoding



Head	Target / loss
NP	Nucleus foreground Cross-entropy + Dice
HV	Normalized horizontal / vertical offsets to nuclear centroid MSE + gradient MSE
NC	Pixel-level nuclear type Cross-entropy + Dice

Sobel derivatives of HV maps expose transitions between touching nuclei. NP supplies the foreground mask and markers. Watershed yields instances, then NC votes assign their types.

UNI provides pathology-pretrained Transformer features

UNI encoder	Specification
Backbone	ViT-L/16
Pretraining	DINOv2 self-supervision
Pretraining data	100M+ patches, 100K+ WSIs
Transformer depth	24 blocks
Embedding dimension	1,024
Model scale	About 307M parameters

Tokenization

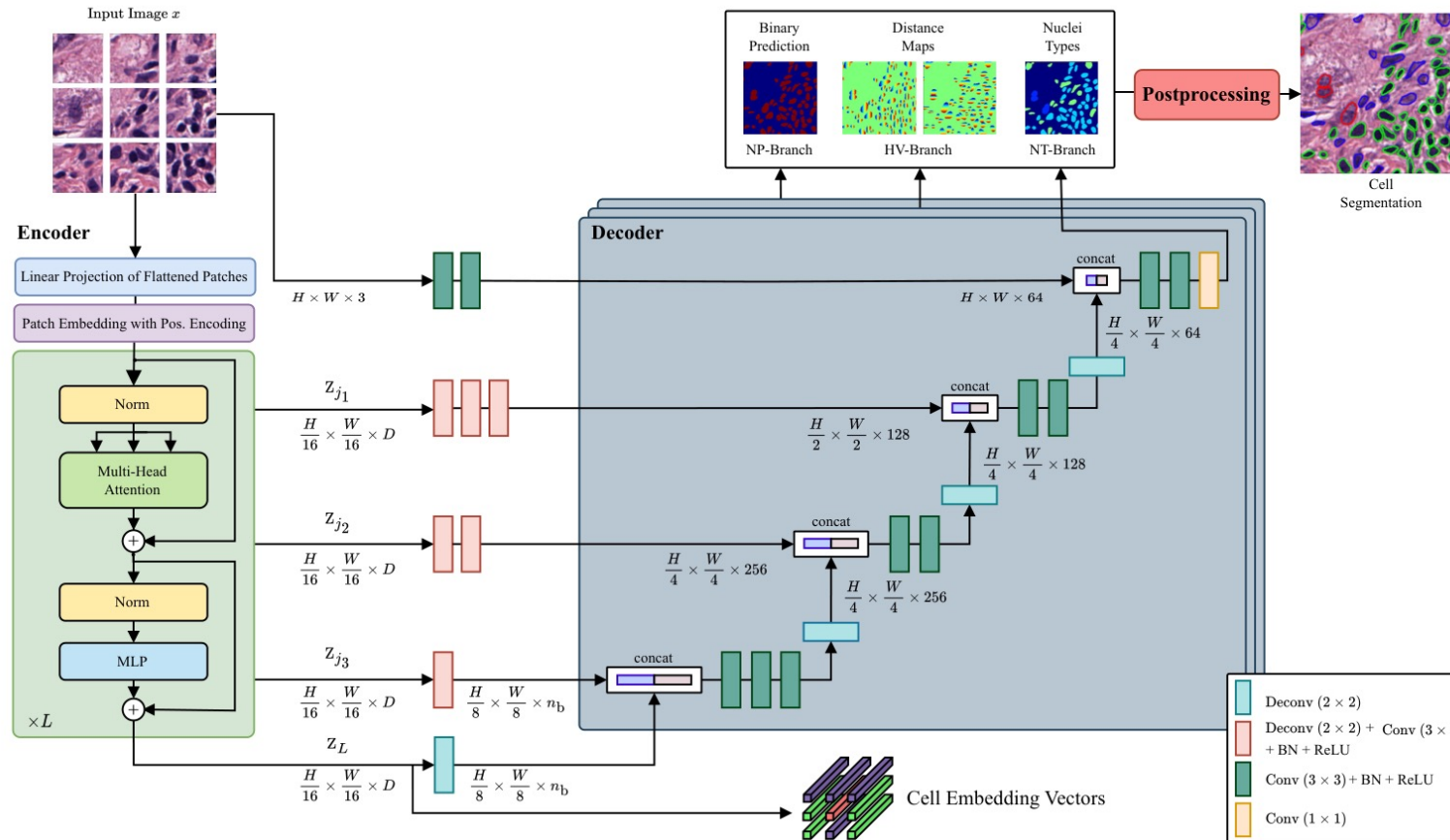
A 256×256 patch produces a 16×16 spatial token grid with 16×16 -pixel tokens.

Dense prediction

Intermediate token features feed a U-shaped decoder. Image-level pretraining still needs a pixel-level segmentation model.

UNI is the image encoder. CellViT adds the nuclear segmentation task

A U-shaped decoder preserves the HoVer output representation



Paper training combines segmentation losses for NP / NT with HV regression. HoVer-style postprocessing separates instances and assigns types.

CellViT-UNI improves mPQ, while Dead remains weak

Model	mPQ	bPQ	Dead PQ
HoVer-Net	0.4564	0.6613	0.127
CellViT-UNI	0.4995	0.6654	0.176
HoVer-NeXt-T	0.4579	0.6367	0.136

3-split means, no TTA. Shared evaluator; validation-only decode thresholds.
HoVer-Net / CellViT-UNI: our training. HoVer-NeXt-T: released per-fold weights.

Observed difference

CellViT-UNI gains 0.0431 mPQ over HoVer-Net, while bPQ differs by only 0.0041.

Persistent deficit

Dead PQ stays at 0.127–0.176 across the three evaluators.

Dead nuclei are lost mainly at detection

GT class	Matched	No overlap	Type accuracy when matched
Neoplastic	81.1%	9.2%	90.5%
Inflammatory	84.9%	7.9%	81.7%
Connective	75.2%	15.3%	79.4%
Dead	47.0%	38.5%	69.0%
Epithelial	83.7%	7.5%	93.0%

Foreground head

Mean NP probability inside missed
Dead nuclei: 0.164
NP activates in only 17% of them.

Type head

$P(\text{Dead}) \approx 0.26$ inside those nuclei
versus ≈ 0.01 in missed nuclei
from other classes.

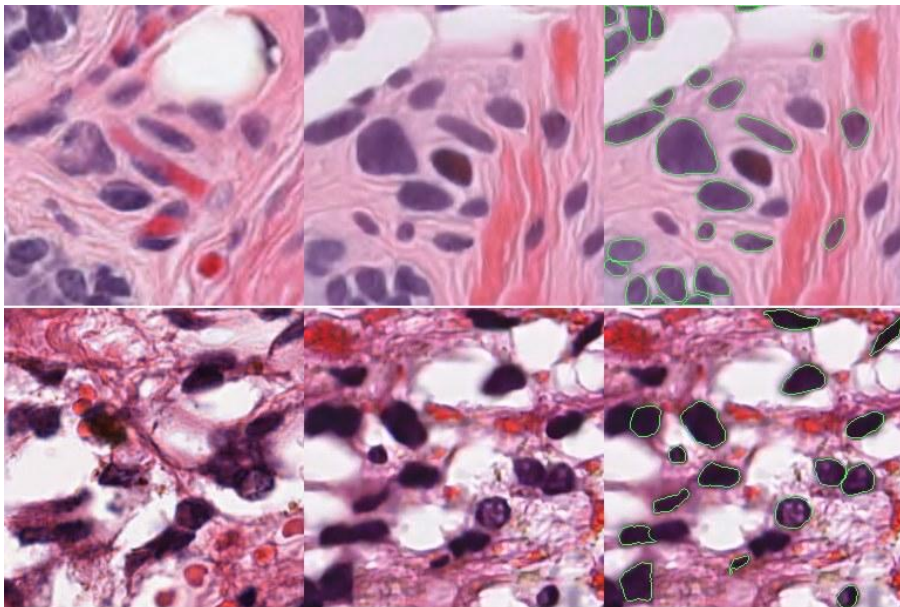
The type signal survives in some missed nuclei, but the foreground gate drops them

Targeted interventions did not establish a reliable gain

Intervention	Evidence	Verdict
CONCH / output recovery	Re-typing gives no reliable gain. 96 decode settings stay within ± 0.0004 mPQ.	Null
Dead weighting / sampling	Dead-pixel weight $\times 11$ and balanced sampling do not help.	Null
Small-nucleus weighting	3-seed Δ mPQ -0.0010 95% CI $[-0.0043, +0.0022]$	Trade-off
Failure-driven copy-paste	3-seed Δ Dead PQ -0.0310 95% CI $[-0.0576, -0.0100]$	Harmful

No tested intervention delivered a convincing gain under controlled evaluation

PixCell preserves layouts but cannot target Dead appearance



Appearance calibration

20× generation differs from 40× PanNuke.
LoRA trials worsened image metrics.
Reinhard transfer reduced OD-L1
from 0.072 to 0.009.

Conditioning blocker

Binary masks and context carry no class label.
0 of 729 kept objects were teacher-typed Dead.
SYN1 used teacher-consistent labels instead.

Source columns: real image / generated image / labels

1,146 images, 50% mixing: test mPQ 0.4900 versus 0.4951. No reliable gain

Zero-shot evaluation separates the architectures

Model	CoNIC mPQ	MoNuSAC mPQ	PUMA mPQ
CellViT-UNI	0.3351	0.2495	0.4518
HoVer-Net	0.2610	0.0370	0.3203
HoVer-NeXt-T	0.2786	0.1123	0.3813

External data

CoNIC: 19,476 tiles, 112 PanNuke patches excluded

MoNuSAC test: 443 tiles

PUMA: 3,278 tiles, all five mapped classes

Failure mechanism

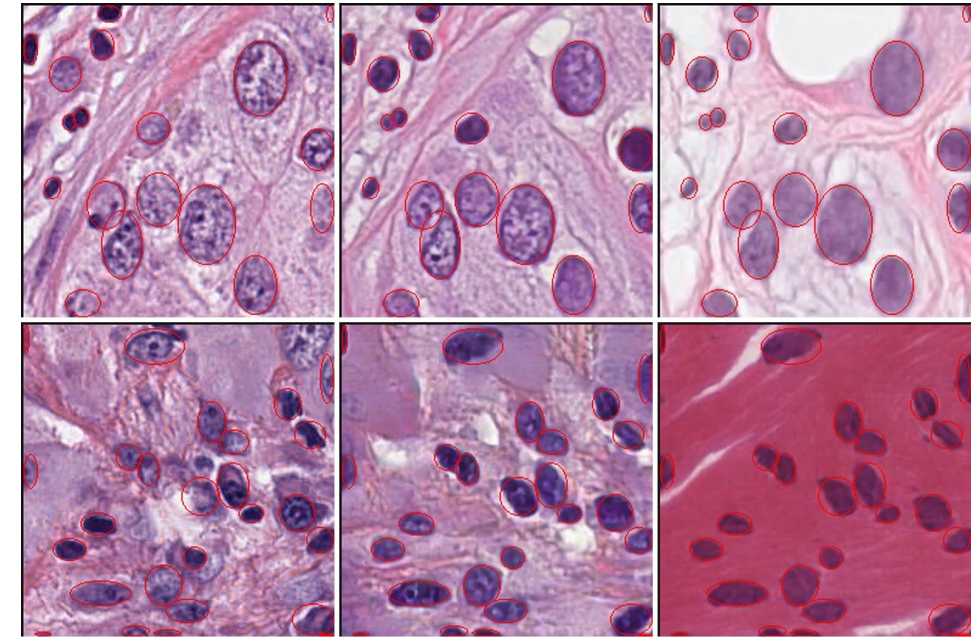
HoVer-Net on MoNuSAC: $F_d = 0.3658$

PUMA detection remains strong: $F_d \approx 0.85\text{--}0.88$

Typing causes much of the PUMA loss.

Fixed labels isolate stain and tissue-context interventions

Arm	Context embedding	Reinhard target
Control	Paired tissue	Self
Stain	Paired tissue	Donor
Context	Swapped tissue	Self
Tissue	Swapped tissue	Donor



Same outlines, different generated appearances

Calibration

500 fold-3 patches. 3 control generator seeds.
Control mPQ SD 0.002–0.006. Exclude blank patches.

Compare each intervention with generated control, not with the real image

Context stress reverses the real-transfer ranking

Model	Control mPQ	Context mPQ	CF context drop	Real domain drop
CellViT-UNI	0.449	0.259	0.190	0.154
HoVer-Net	0.440	0.300	0.140	0.246
HoVer-NeXt-T	0.425	0.262	0.163	0.199

Observed inversion

CellViT-UNI loses most under CF context swap, yet loses least on the real external datasets.

Limited statistical scope

6 evaluator points: Spearman $\rho = -0.89$
3 architecture means: $\rho = -1.00$
Only 3 independent architectures.

Appearance counterfactuals mainly probe typing

Model	Real Fd drop	CF Fd drop	Real typing gap increase	CF typing gap increase
CellViT-UNI	0.007	0.102	0.088	0.082
HoVer-Net	0.163	0.084	0.069	0.034
HoVer-NeXt-T	0.045	0.090	0.085	0.064

Real transfer

Architecture differences concentrate on detection. HoVer-Net collapses on MoNuSAC while CellViT-UNI retains detection.

Appearance counterfactuals

Fixed layouts re-render well-formed nuclei. Detection drops are similar. Typing ranks agree with real transfer.

Useful as a typing stress test. Insufficient as a general transfer surrogate

Phase I is complete. The next method line targets detection

1

Completed

Shared baselines and error analysis. Pillars A / B closed with null or harmful results. Pillar C yielded a benchmark-validity result.

2

Next experiments

Strictly re-evaluate released detection-first models. Compare 2× input resolution under the same validation-only decode protocol.

3

Final report

Complete tissue × class heatmaps and error figures. Add border-aware and Uterus-excluded Dead reporting.

Next: test detection-first modeling and resolution under the same strict protocol